Fragmentation of a Gravito-Turbulent Disc on an Adaptive Mesh in the AthenaK Shearing Box

Test Results

the clump at an interface, mass renormalization, $\beta = 10$, 5 and 3, Jeans-number AMR, a fragment followed to 64 cells/H, and α' against β

`athenak-multigrid` tree, branch `dust-multigrid`

runs on NAS (Turin nodes), TACC Vista and the UNLV h-cluster

September 28 – November 5, 2026

Abstract

This document records what the code described in `sc14_fragmentation_implementation.pdf` does on its test problems and in the production runs of the Shi & Chiang (2014, SC14) stratified, self-gravitating, β-cooled shearing box. All production runs use the $64H \times 64H \times 16H$ box at 4 cells/H on the root grid. Where a run is still going, its state on October 15 is given and marked.

Test problems. A bound clump next to a coarse–fine interface drifts toward the coarse side under the multigrid force error, $\sim 2 \times 10^{-3}$ of its surface gravity; a gap of 8 fine cells cuts the drift 60×, which sets the refinement buffer. For a moving clump WENO-Z with flux correction and PPMX track the reference best, PPM4 worst.

The box. The $16H$-tall box loses 10% of its mass by $t = 300$ (the $12H$ box 24%); renormalization holds the mass exactly and leaves Q unchanged. At $\beta = 10$ nothing fragments — not in four uniform runs to $t = 300$, not in a Jeans-AMR run to $t = 1000$. At $\beta = 5$ (Jeans AMR, running) nothing has fragmented in $110/\Omega$. At $\beta = 3$ the WENO-Z run fragments at $t \approx 313$ and keeps the fragment to $t = 400$; plm only makes transient knots.

Stress against cooling. Over $\beta = 4$–80 the WENO-Z+FOFC stress matches the thermal balance (SC14 eq. 21) to $\alpha'/\text{eq. } 21 = 1.01$–$1.07$, and the energy budget closes once the energy leaving through the vertical boundaries (0.06–0.17 of the cooling) is counted. plm falls short at high β (0.90 at $\beta = 40$, 0.82 at 80, with negative Reynolds stress); PPM4 overshoots at $\beta = 80$ because its disc loses energy and mass through the boundaries (§5).

The fragment followed. With refinement that follows it, the $\beta = 3$ fragment contracts on the cooling time without limit ($\rho_{\max}$ doubles per Ω^{-1}, $687 \to 4 \times 10^5$ from $t = 316.8$ to 326.85), pulling levels 4 and 5 in behind it until the run slows to $0.08/\Omega$ per hour. Its outer structure ($R > 0.5H$) matches the uniform run; inside, half its mass sits within $0.25H$ against 8% on the uniform grid, and it spins up: peak v_ϕ/v_K rises from 0.45 to 0.83 while the densest gas flattens about 4–5:1. By the 45%-Keplerian criterion of Szulágyi et al. (2017) it has a disc of ~ 330 mass units, about half its Hill mass, whose inner edge is unresolved. A first-core cutoff in the cooling now stops the contraction; that run is in progress.

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1 Runs

run (NAS dir. unless noted)	job	setup and state
`gt_sc14_amr_b10_restart_L2` (Vista)	—	$12H$ box, 4 cells/H, one refined level, $\beta = 10$ from $t = 0$ (archived run restarted); done to $t = 300$
`b10`, `b10_renorm`	244225/6	$64H$ box, WENO-Z+FOFC, $\beta = 10$, $t = 0$–300, renormalization off/on; done
`b10_plm`, `b10_plm_renorm`	244234/5	the same with plm, no FOFC; done
`b3_renorm`, `b3_plm_renorm`	244909/8	$\beta = 3$ from the renormalized $\beta = 10$ runs at $t = 300$, to 400; done
`b10_renorm_amr`	245022…	Jeans AMR to level 3 (32 cells/H), `amr_njeans` 16, $\beta = 10$, $t = 100$–1000; done
`b5_renorm_amr`	248988	the same at $\beta = 5$ from $t = 300$ (cap 256 per rank); *running*, $t = 412$ on October 15
`b3_renorm_clumpamr_nj16_crashed`	245052	fragment-following AMR, $\beta = 3$ from $t = 300$, `amr_njeans` 16, `amr_cs_min` 0.3; crashed $t = 307.8$
`b3_renorm_clumpamr`	245227, 247997	the same with `amr_njeans` 8, `amr_cs_min` 1, hold $\rho > 100$ at level 3, buffer; $t = 300$–326.85
`b3_renorm_clumpamr_fc`	250762	continued from $t = 324.94$ with `bcool_rho_cut` $= 10^5$, deepest level 4; *running*
`b{20,40,80}lad_renorm_r4`	259985–7	WENO-Z ladder from the β-ramp run `b10ramp_renorm_r4` at $t = 100$: $\beta = 20$ to $t = 250$, then 40 to 550, then 80 to 950; dumps every $2/\Omega$; done (November 2026)
`b{5,4}lad_renorm_r4`	260485/6	$\beta = 5$ and 4 directly from the same restart, $t = 100$–300; done

All $64H$-box runs: $256 \times 256 \times 64$ root grid, 8^3 MeshBlocks, 512 ranks, WENO-Z+FOFC+HLLC+`rk2` unless noted, `hse_outflow` in z, dumps every $1/\Omega$ (the fragment-following runs every $0.1/\Omega$). $\Sigma_0 = 2.0026$ is the mean initial surface density; α' is SC14 eq. 20, and eq. 21 predicts $\alpha' = 4/[9\gamma(\gamma-1)\beta]$ (0.040 at $\beta = 10$, 0.080 at 5).

2 A clump at a coarse–fine interface

2.1 The interface ladder

`sg_clump`: an $n = 1$ polytrope ($R = 1$, $\rho_c = 1$, $4\pi G = 1$, mean free-fall time 3.5) in a $\pm 4R$ periodic box whose half $x > 0$ is refined by one level ($R/\Delta x = 8$ coarse, 16 fine; plm), with the centre at x_0 relative to the interface, to $t = 30$ (~ 9 free-fall times). Uniform coarse and fine twins bracket it.

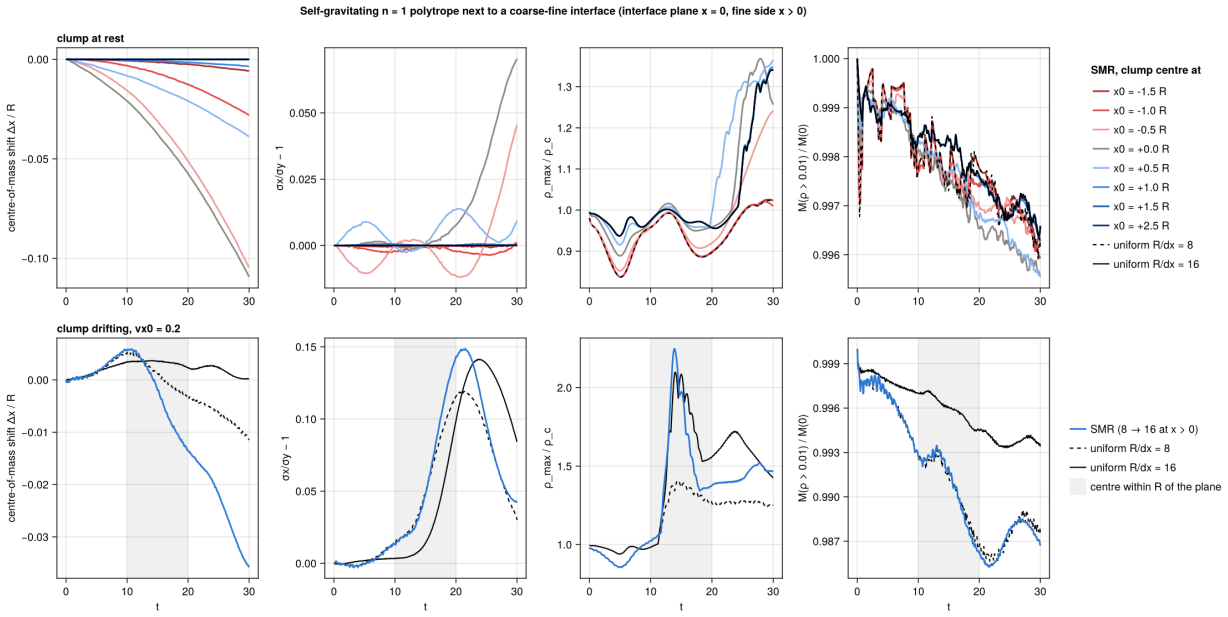


Figure 1: Clump at a coarse–fine interface: centre-of-mass drift, spurious acceleration, aspect ratio and peak density against the position of the centre relative to the interface plane, with the uniform twins.

- The clump always drifts **toward the coarse side**, with a nearly constant acceleration $\sim 2 \times 10^{-4}$ ($\sim 2 \times 10^{-3}$ of its surface gravity GM/R^2): $\Delta x(t = 30) = -0.105R$ ($x_0 = -0.5$), $-0.109R$ ($x_0 = 0$), $-0.039R$ ($x_0 = +0.5$).
- Its x/y shape grows by 4.5–7% after $t \sim 20$; mass and peak density follow the side the centre is on. No disruption.
- A clump at least $0.5R$ (8 fine cells) inside the fine side is unaffected (10^{-4}): the gap cuts the drift 60×.
- A clump crossing the plane ($v_{x0} = 0.2$) loses $\sim 1\%$ of its velocity.

Consequence. The multigrid force error at a level jump is real but small, and it is local: keeping fragments 8 fine cells from an interface removes it. The refinement *buffer* (implementation document) provides exactly that.

2.2 Reconstruction for a collapsing clump

The same clump, uniform $R/\Delta x = 8$, at rest and drifting, with plm at $R/\Delta x = 16$ as the reference.

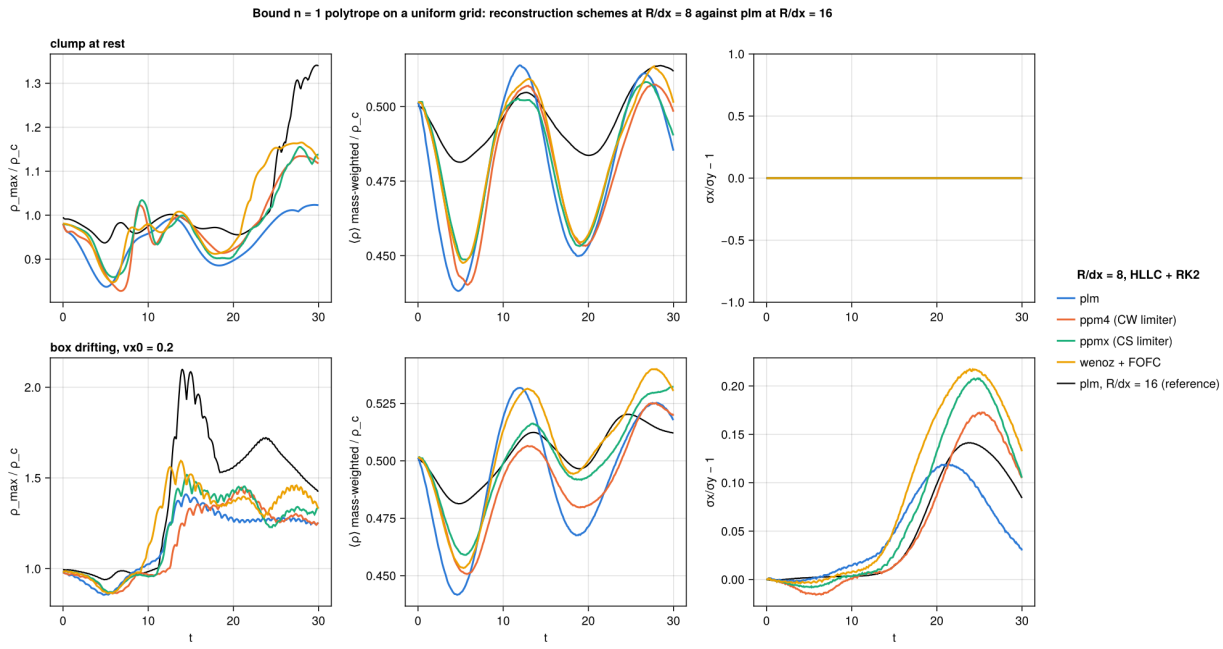


Figure 2: Reconstructions on the clump, at rest and drifting, against the plm $R/\Delta x = 16$ reference.

At rest PPM4, PPMX and WENO-Z+FOFC behave alike and stay closer to the reference than plm. Drifting, PPM4 is the most diffusive of the high-order schemes (its Colella–Woodward limiter clips the moving extremum and its mass-weighted density tracks plm); PPMX and WENO-Z+FOFC stay closest; WENO-Z overshoots the elongation most. The drifting runs show a compression event at $t \sim 11$–14 ($\rho_{\max} \times 1.5$–2.1) absent at rest, larger at higher resolution and unexplained (a Galilean-invariance artefact of the setup). WENO-Z+FOFC costs 15–20% more than PPM4 here, 5% in the SC14 box. In the $12H$ SC14 box at $\beta = 10$ the three schemes give the same saturated state: α'/eq. 21 = 0.997 (plm), 1.023 (PPM4), 0.990 (WENO-Z+FOFC), within the ± 0.12–0.15 scatter. **Choice:** WENO-Z+FOFC for the production runs.

3 One refined level in the 12H box ($\beta = 10$, Vista)

The archived 4 cells/H run restarted at $t = 0$ with one refined level (8 cells/H) and the Jeans criterion, to $t = 300$.

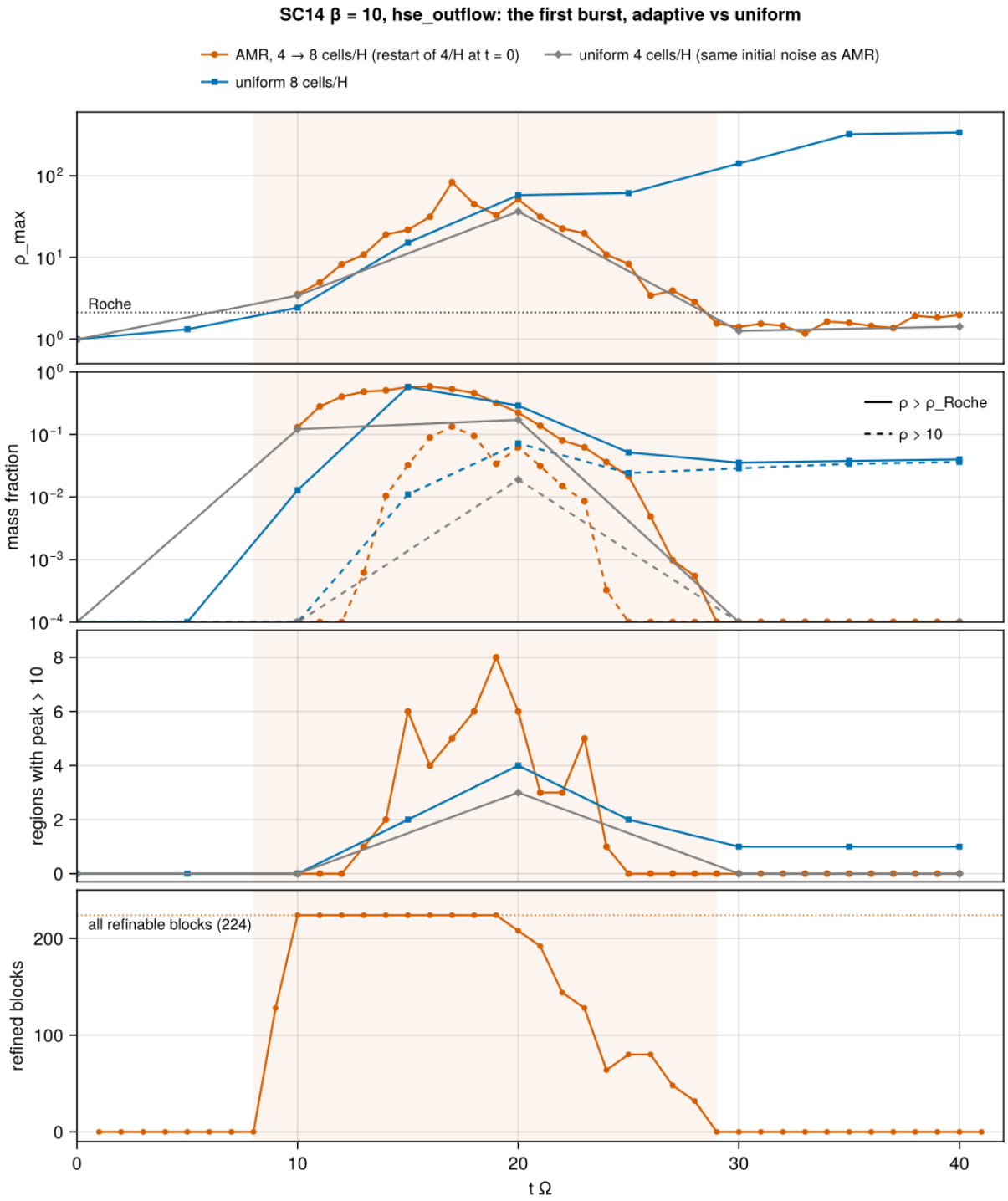


Figure 3: Clump census of the one-level AMR run against the uniform 8 cells/H run.

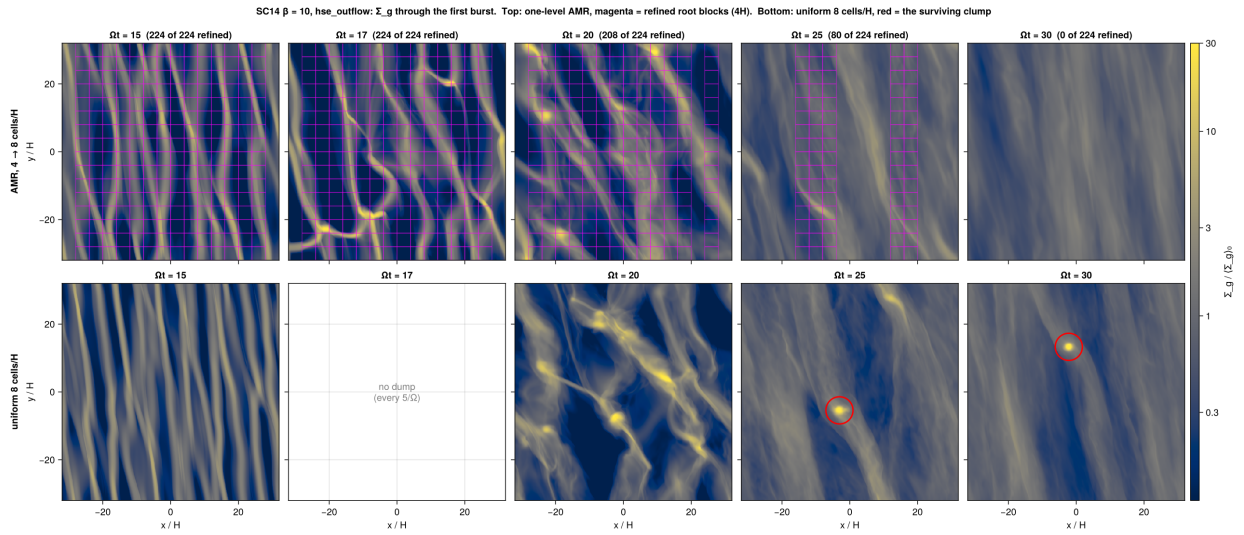


Figure 4: Σ_g through the first burst, refined MeshBlocks marked (top), and the uniform 8 cells/H run (bottom).

The criterion refined the whole refinable midplane (all 224 parents, 2336 blocks, the layout maximum $|z| < 2H$, $|x| < 28H$) from $t = 10$ to 19, then let it go by $t = 29$. The burst reached the uniform hi-res densities ($\rho_{\max} = 22$ at $t = 15$, 83 at 17, 51 at 20, up to 6–8 knots above 10) but every knot decayed *while still on level 1*; nothing was above the Roche density after $t = 29$. The uniform 8 cells/H run kept one fragment of the same burst (141 at $t = 30$). So derefinement did not destroy the knots. A four-level run of the same setup died twice at $t \approx 15$ in MPI (UCX "transport retry count exceeded"); unresolved, on hold.

4 The $64H$ box at $\beta = 10$: mass renormalization and reconstruction

Four runs, $t = 0$–300: WENO-Z+FOFC and plm, each without and with renormalization.

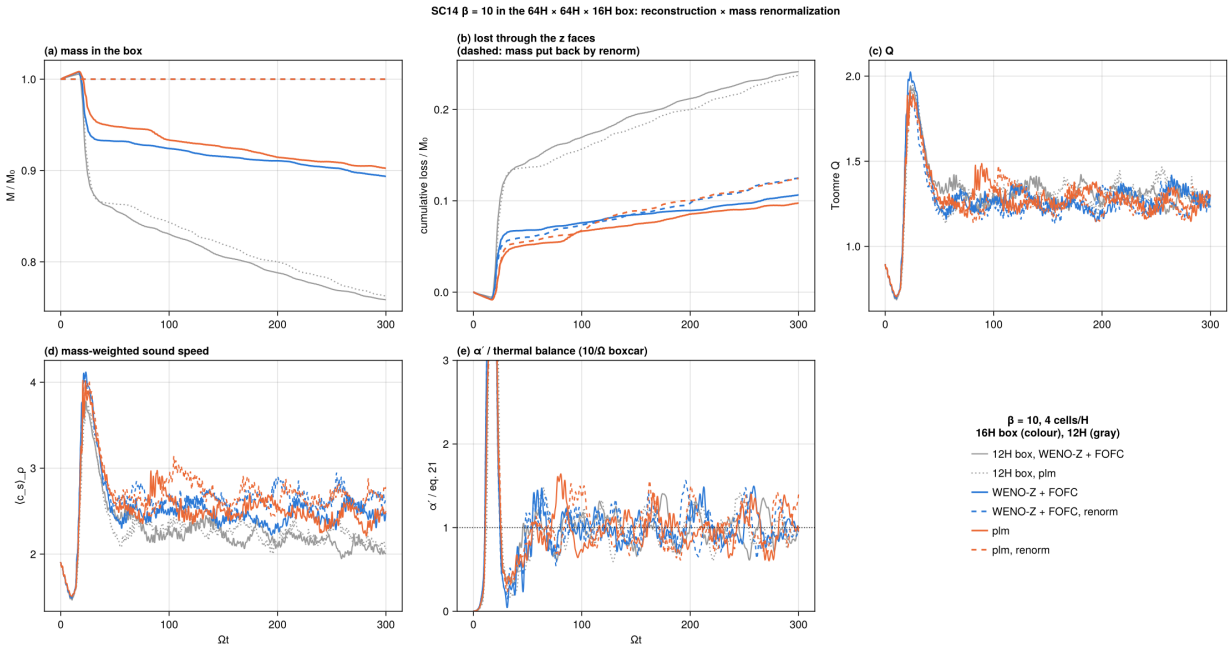


Figure 5: Histories of the four $\beta = 10$ runs: box mass, cumulative mass loss, α', Q.

- **Saturated state** ($t = 100$–300): $Q = 1.24$–1.27 in all four (the $12H$ box 1.30–1.32); α'/eq. 21 $= 0.99$–1.01. WENO-Z and plm agree within the scatter (WENO-Z bursts $1.5/\Omega$ earlier, peak α 0.53 against 0.40).
- **Mass loss.** $M(300)/M_0 = 0.894$ (WENO-Z) and 0.902 (plm), against 0.759 and 0.762 in the $12H$ box: the burst takes $\sim 6\%$ ($12H$: 14%), then 1.5×10^{-4} per Ω^{-1} ($12H$: 3.9×10^{-4}).
- **Renormalization** holds $M = 8202.52$ to the printed precision and puts back 12.5% of M_0 by $t = 300$ (5.5% in the burst, then 2.8×10^{-4} per Ω^{-1}). It does *not* change Q: c_s rises 7–9% with the retained Σ (the disc self-regulates), δv by 8%.

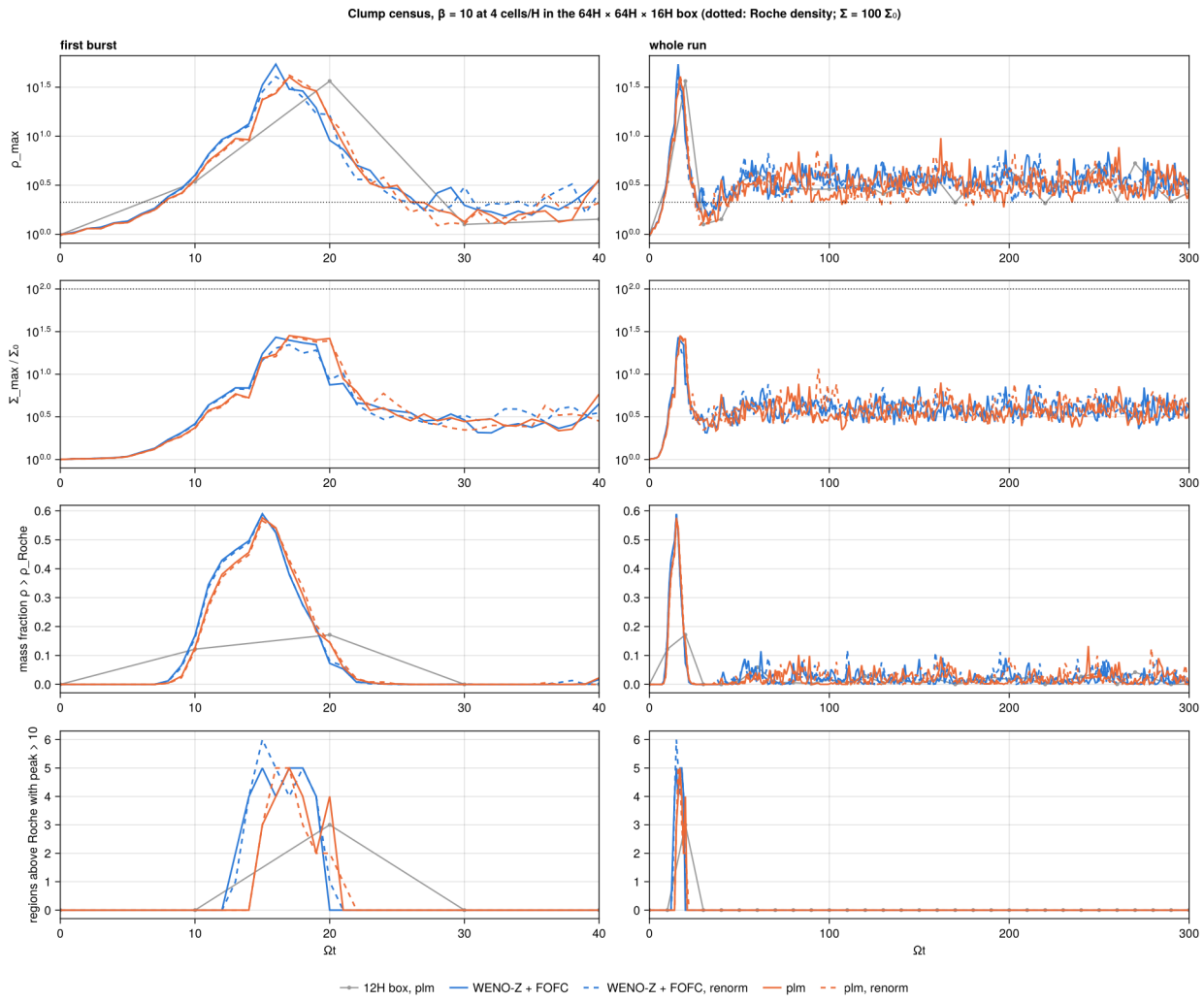


Figure 6: Clump census of the four $\beta = 10$ runs: $\rho_{\max}$, $\Sigma_{\max}/\Sigma_0$, mass above the Roche density, knots above Roche with peak > 10. Gray: the $12H$ box.

No fragment at $\beta = 10$, 4 cells/H. The burst reaches $\rho_{\max} = 40$–54 at $t = 16$–17 (WENO-Z highest), $\Sigma_{\max}/\Sigma_0 = 22$–$28$, 4–6 knots, all gone by $t \approx 20$–22. After $t = 40$: $\rho_{\max} \leq 9.6$, $\Sigma_{\max}/\Sigma_0 \leq 11.6$ (Paardekooper's criterion is 100), never a knot above 10. Neither renormalization nor reconstruction changes this.

4.1 Would Jeans AMR have switched on?

The refinement criterion applied to the uniform dumps (`gt_jeans_census.py`, counts as full-y rings):

- **Burst**: $n_J < 16$ (level 1) from $t = 8$ to 22, over the whole midplane (all 64 $|z| < 2H$ columns at $t = 10$–18); $n_J < 8$ (level 2) at $t = 14$–19 (≤ 24 rings); never $n_J < 4$ (min 5.4 and 5.1).
- **Saturated state** ($t \geq 40$): $n_J < 16$ in only 12 of 261 dumps (`b10`) and 1 of 261 (`b10_renorm`, the hotter disc), minimum 14.4, 1–5 rings. The dense gas is hot; the median of the per-dump minimum is 18–20.

- $\beta = 3$ (WENO-Z): the fragment has $n_J \approx 3.5$–5 at 4 cells/H (needs level 3); cold streams reach $n_J \approx 2$ with a sound-speed floor of 0.3. With `amr_njeans` 16, 30–58 rings would be at level 1 before the fragment forms; with 8, at most one.

5 α' against β: reconstruction and the cooling balance

In a steady gravito-turbulent state the stress heats the disc as fast as β cooling removes energy, which fixes $\alpha' = 4/[9\gamma(\gamma-1)\beta]$ (SC14 eq. 21). A first ladder with plm reconstruction ($\beta = 10 \to 20 \to 40 \to 80$, each stage restarted from the end of the previous one) fell short of it at large β. This section reruns the ladder with PPM4 and with WENO-Z+FOFC and closes the energy budget, to separate the scheme from the physics.

Runs.

- **plm and PPM4 ladders** (UNLV h-3; `validation/run/gt_sc14_stndrd_b*_hse{,_ppm4}`): the $64H \times 64H \times 12H$ SC14 box at 4 cells/H, `hse_outflow`, no mass renormalization; $\beta = 10$ for $t = 0$–300, then 20 to 450, 40 to 750, 80 to 1150. Also WENO-Z+FOFC at $\beta = 10$ (`gt_sc14_stndrd_b10_hse_wzfofc`).
- **WENO-Z+FOFC ladder** (NAS): the production $64H \times 64H \times 16H$ box at 4 cells/H with mass renormalization, restarted from the β-ramp run (`b10ramp_renorm_r4`, β from 100 to 10 by $t = 50$, then 10 to $t = 100$): $\beta = 20$ to $t = 250$, then 40 to 550, then 80 to 950; and $\beta = 5$ and 4 directly from $t = 100$ to 300. Dumps of $\rho, \mathbf{v}, P$ and Φ every $2/\Omega$.

α' (SC14 eq. 20, from the history every $0.25/\Omega$) is averaged over the final window of each stage: $t = 200$–300, 350–450, 550–750 and 950–1150 for the standard box; 150–300 ($\beta = 4$, 5), 75–100 ($\beta = 10$), 150–250, 350–550 and 750–950 ($\beta = 20$, 40, 80) for the production box. The errors are the standard error of $5/\Omega$ block means (correlated, so approximate).

Energy budget. `scripts/analysis/gt_energy_budget.py` integrates the total energy $E = K + U + E_{\rm tid} + E_{\rm sg}$ of the gas in the box over paired dumps,

$$\frac{dE}{dt} = S - C - F_z + R,$$

with $S = q\Omega \int w_{xy}\, dV$ the stress work, $C = \int U/\beta\, dV$ the cooling, F_z the energy carried out through the two z faces (kinetic, enthalpy, tidal and self-gravitational potential), and R a residual for floors, the non-conservative self-gravity coupling and the renormalization. In a steady state $S - F_z \approx C$.

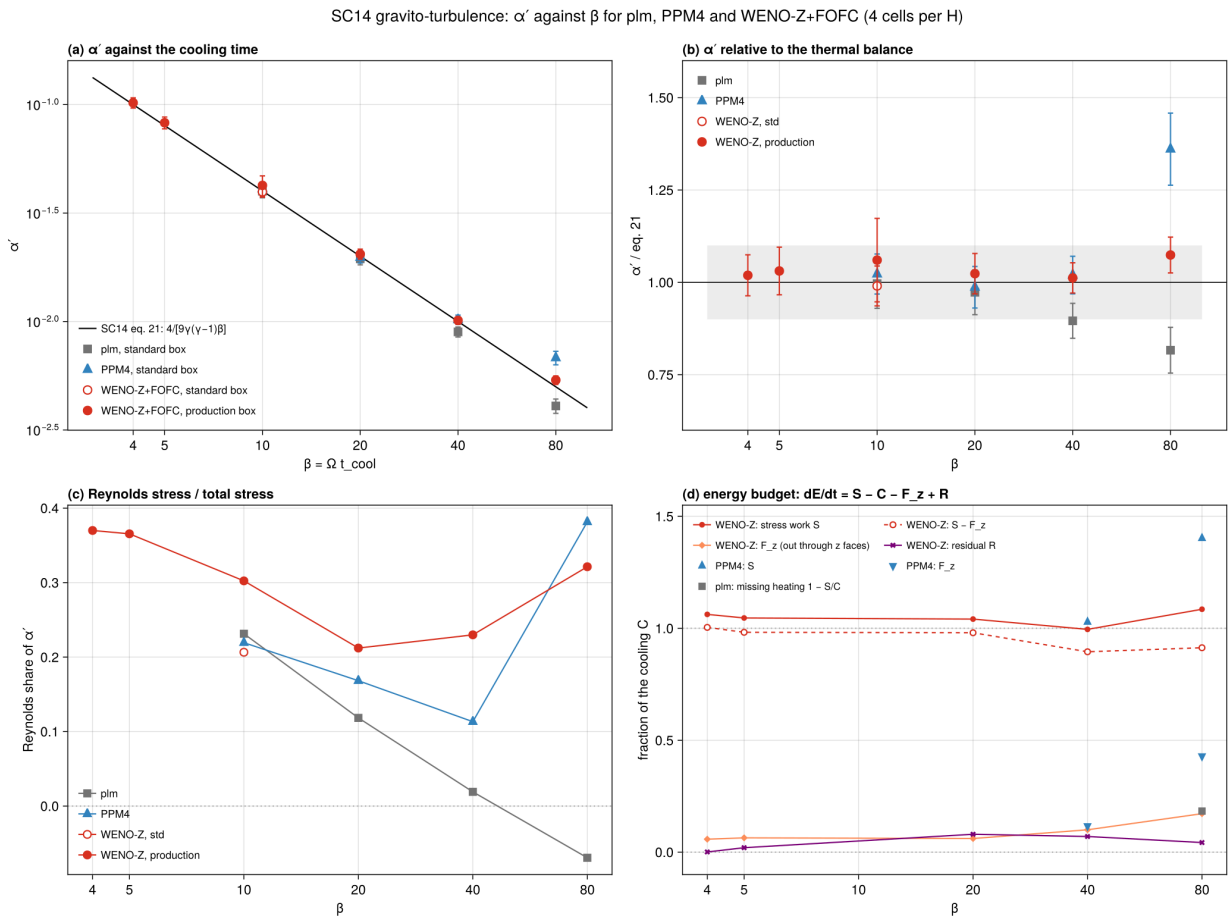


Figure 7: α' against β. (a) α' with SC14 eq. 21; (b) $\alpha'/\text{eq. 21}$ with a $\pm 10\%$ band; (c) the Reynolds share of α'; (d) the WENO-Z energy budget as fractions of the cooling C (stress work S, $S - F_z$, the boundary loss F_z, the residual R), with PPM4 at $\beta = 40$, 80 and the heating plm lacks at $\beta = 80$. Squares: plm; triangles: PPM4; open circle: WENO-Z+FOFC in the standard box; filled circles: WENO-Z+FOFC in the production box.

β	eq. 21	plm	PPM4	WENO-Z (std)	WENO-Z (prod.)	Reynolds share (WENO-Z)
4	0.100	–	–	–	1.02 ± 0.06	0.37
5	0.080	–	–	–	1.03 ± 0.06	0.37
10	0.040	1.00	1.02	0.99	1.06 ± 0.10	0.30
20	0.020	0.97	0.99	–	1.02 ± 0.05	0.21
40	0.010	0.90	1.02	–	1.01 ± 0.04	0.23
80	0.005	0.82	1.36	–	1.07 ± 0.05	0.32

Table 2: $\alpha'/\text{eq. 21}$ by scheme ($\pm$: standard error of $5/\Omega$ block means, about 0.03–0.06 for plm and PPM4). The Reynolds share of α' for plm is 0.23, 0.12, 0.02 and -0.07 at $\beta = 10$–80.

run	S/C	F_z/C	$(dE/dt)/C$	R/C	$(S-F_z)/C$
WENO-Z, $\beta = 4$	1.06	0.06	+0.004	+0.001	1.00
WENO-Z, $\beta = 5$	1.05	0.06	+0.003	+0.02	0.98
WENO-Z, $\beta = 20$	1.04	0.06	+0.06	+0.08	0.98
WENO-Z, $\beta = 40$	1.00	0.10	−0.04	+0.07	0.90
WENO-Z, $\beta = 80$	1.09	0.17	−0.04	+0.04	0.91
PPM4, $\beta = 40$	1.03	0.12	−0.04	+0.05	0.91
PPM4, $\beta = 80$	1.40	0.43	−0.02	+0.003	0.97

Table 3: Energy budgets over the averaging windows, as fractions of the cooling C.

- **plm falls short at large β.** $\alpha'/\text{eq. } 21 = 1.00, 0.97, 0.90, 0.82$ at $\beta = 10, 20, 40, 80$ (a fit gives $\alpha' \propto \beta^{-1.10}$), and the Reynolds stress turns negative at $\beta = 80$. Its disc barely loses energy through the boundaries, so about 16% of the cooling at $\beta = 80$ is supplied by heating that the stresses do not measure. (The plm budget was not computed: only the final plm dumps were kept.)

- **PPM4 agrees to** $\beta = 40$ (0.99, 1.02) and overshoots at $\beta = 80$ (1.36). The budget closes to 0.3% of C: the excess stress heating leaves through the z faces ($F_z = 0.43C$: tidal potential 0.15, self-gravity 0.13, enthalpy 0.12, kinetic 0.04), while the disc heats (c_s 2.03 to 2.40) and loses 15% of its mass. The overshoot is the outflow, not an error in the stress.

- **WENO-Z+FOFC matches eq. 21 at every** β from 4 to 80 (1.01–1.07), with a positive Reynolds share of 20–37%. At $\beta = 10$ in the standard box it gives 0.99, the same as plm and PPM4, which ties the two boxes together. The budgets close: F_z grows from $0.06C$ ($\beta \le 20$) to $0.17C$ ($\beta = 80$), the residual stays $\le 0.08C$, and the renormalization keeps the mass (its energy input is $\sim 0.01C$).

- $\beta = 20$ **is still settling**: $dE/dt = +0.06C$ (the disc warming after the switch from $\beta = 10$; the window is five cooling times). $\beta = 4$ and 5 close to 0.1–2% of C.

- **No fragment at** $\beta = 4$ **or 5** on the uniform 4 cells/H grid in $200/\Omega$ (every $2/\Omega$ dump, `sigma_peaks.py`). $\beta = 4$: $\rho_{\max} \le 21.9$ (above 20 in three dumps only, median 8.2), $\Sigma_{\max}/\langle\Sigma\rangle \le 17.7$ (median 7.8); $\beta = 5$: $\rho_{\max} \le 15.3$ (median 6.3), $\Sigma_{\max}/\langle\Sigma\rangle \le 15.6$. Both stay far below Paardekooper's criterion of 100, so their α' is that of the turbulence, not of a bound clump. At $\beta \ge 20$ the peaks fall to $\rho_{\max} \le 3.5$.

The α' deficit at large β belongs to plm. With WENO-Z+FOFC the stress balances β cooling to within 7% from $\beta = 4$ to 80, and the energy budget closes once the boundary outflow is counted. The comparison is between setups as well as schemes (the plm and PPM4 ladders ran in the $12H$ box without renormalization); the WENO-Z $\beta = 10$ run in the $12H$ box agrees with plm and PPM4.

6 $\beta = 3$ on the uniform grid

The renormalized $\beta = 10$ runs, restarted at $t = 300$ with $\beta = 3$.

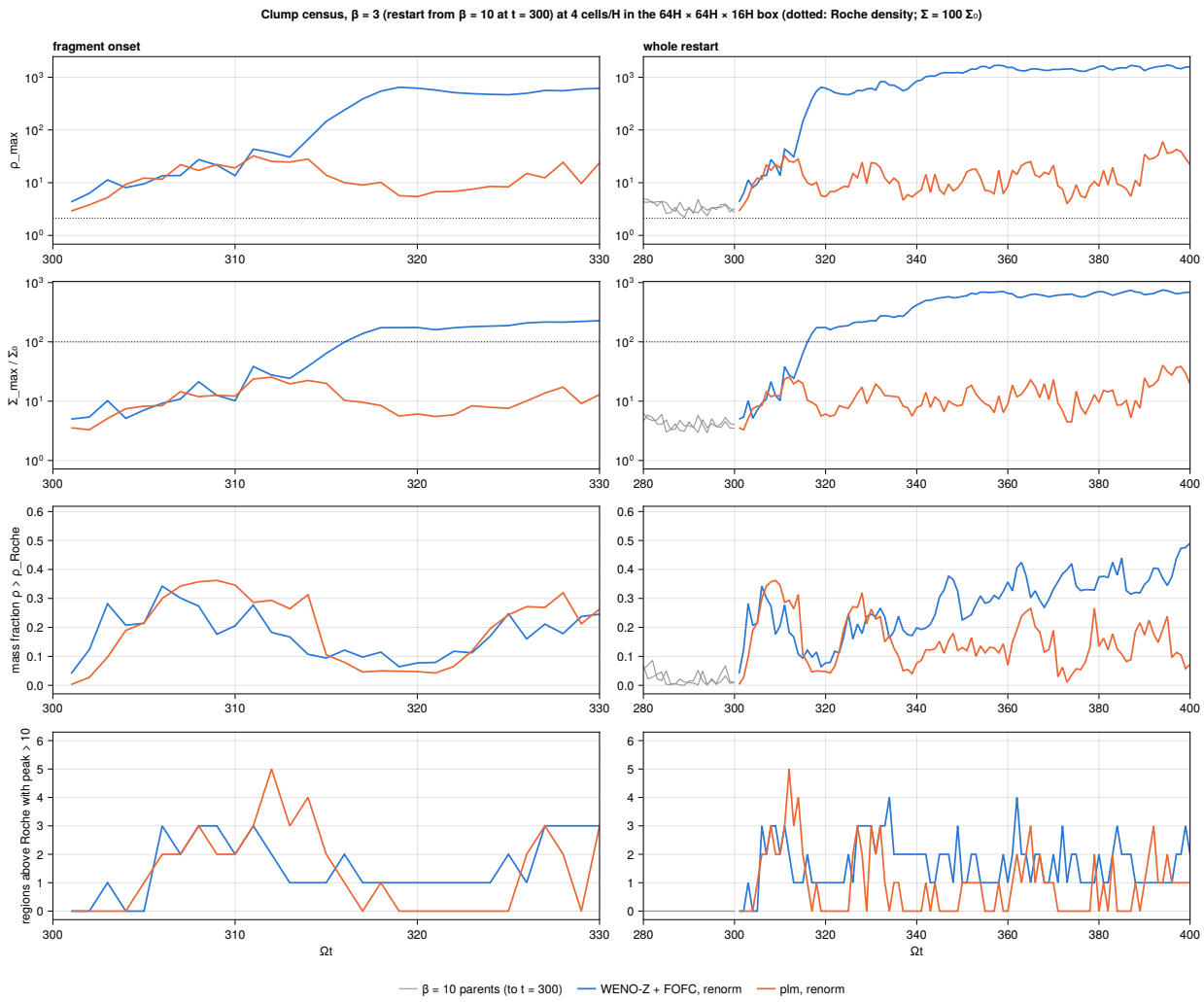


Figure 8: Clump census of the $\beta = 3$ restarts, WENO-Z+FOFC and plm; gray: the $\beta = 10$ parents.

Reconstruction decides fragmentation at this resolution. The two runs track each other to $t \approx 313$ and part there. WENO-Z+FOFC fragments: $\rho_{\max}$ goes from 30 to 600 by $t \approx 319$ and stays at 1500–1700 from $t \approx 355$ to 400; $\Sigma_{\max}/\Sigma_0$ is above 100 from $t \approx 316$ (175 at $t = 320$, ~ 700 at $t = 380$–400), for $85/\Omega$ — many cooling times. Plm never exceeds $\rho_{\max} \approx 60$ or $\Sigma_{\max}/\Sigma_0 \approx 40$; its knots dissolve within a few Ω^{-1}.

The WENO-Z fragment accretes steadily (mass above the Roche density 0.2 → 0.49 of the box by $t = 400$), clears a wide low-density band around itself, and moves outward in x (to $x \approx 16$ by $t = 400$). At $t \approx 333$–343 it jumps $\sim 10H$ outward within $3/\Omega$ while the mass in its Hill sphere doubles (630 → 1200); no second fragment above $30\Sigma_0$ is involved.

7 Jeans-number AMR at $\beta = 10$ and 5

7.1 $\beta = 10$, $t = 100$–1000

Jeans AMR to level 3 on the renormalized WENO-Z run from $t = 100$. Refinement is almost never needed: level-1 patches of 256 blocks at isolated times ($t = 351$, 720); every dump sampled every 20–30/Ω from $t = 440$ to 1000 is root only, with $\rho_{\max} = 2$–8. Time-step dips below 60% of the median occur 1.5% of the time, as often as in the uniform run — hot spots, not refinement.

	$t = 100$–400	400–700	700–1000
α'	0.038	0.037	0.039

against 0.040 from eq. 21; the mass stays at 8202.52.

A 900/Ω null result. At 4 cells/H with Jeans refinement standing by, $\beta = 10$ does not fragment between $t = 100$ and 1000.

7.2 $\beta = 5$ from $t = 300$ (running)

The same refinement, β lowered to 5 at $t = 300$. As the disc cools to its new state, level 1 switches on at $t \approx 302$ and then waxes and wanes (768 to 13 000 level-1 blocks, up to ~ 45 of the 256 x–z columns); level 2 has never been needed; $\rho_{\max} \leq 16$. $\alpha' = 0.068$–0.084 (0.080 for $t = 360$–413), as eq. 21 predicts. The disc reorganizes into broad trailing bands with wide gaps. About 5/Ω per hour.

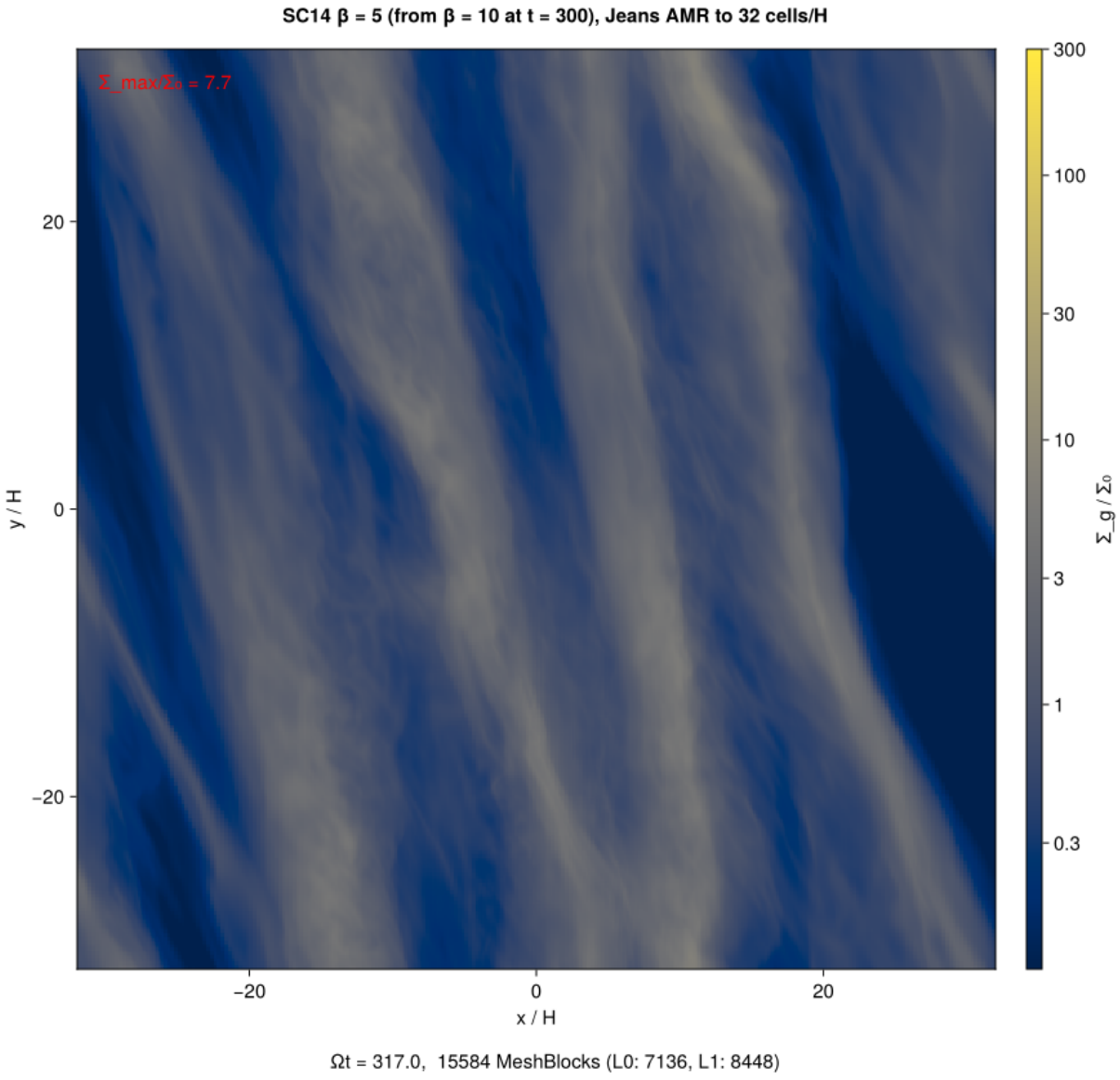


Figure 9: $\beta = 5$ AMR run at $t = 317$: Σ_g/Σ_0 (no mesh drawn); 7136 root and 8448 level-1 blocks.

State on October 15. No fragment in $110/\Omega$ at $\beta = 5$, between $\beta = 3$ (fragment within $14/\Omega$) and $\beta = 10$ (none in $900/\Omega$).

8 Following the fragment ($\beta = 3$)

8.1 First attempt: the cold streams

`amr_njeans` 16, `amr_cs_min` 0.3, up to level 6, hold at level 3 above $\rho = 100$. Level 1 appeared at $t \approx 302$ (13–23k blocks); by $t = 307.1$ the run was at level 4 with 70–104k blocks (~ 200 per rank of 256) while $\rho_{\max}$ was only 26–45: the cold streams ahead of the fragment, not the fragment, drove it. The time step fell to $1.4\text{–}3 \times 10^{-4}$ at ~ 5 s per cycle ($t = 400$ would have taken ~ 400 h).

At $t \approx 307.8$ the run died of glibc heap corruption, not at a dump, with 331 GB of 1.5 TB in use (implementation document, known limitations). No restart file after $t = 300$ (they were every $10/\Omega$).

8.2 Second attempt: refine only what collapses

`amr_njeans` 8 (twice Truelove's 4), `amr_cs_min` 1, restart files every $1/\Omega$, otherwise as before.

t	blocks	level counts (L0/L1/L2/L3/L4/L5)	$\rho_{\max}$
300–307	8192	root only	3–13
308–311	9.5–11.3k	level-1 patches on the streams	14–54
312–313	8192	root only again	35
314.2	9088	8064/1024	101 (hold)
314.4	16 704	8000/1152/2432/5120	106
316.8	24 096	7904/1728/3200/11264	687
318.8	25 440	7840/2112/4224/11264	2942
320.8	20 736	7936/1536/3072/8192	11 754
321.8	35 744	7904/1728/3072/10752/12288	22 558
323.8	47 840	7904/1536/4352/11520/22528	81 254
324.8	53 888	7808/2304/4352/10752/28672	1.28×10^5
326.1	73 376	7776/2752/3456/6144/20480/32768	2.24×10^5
326.85	111 008	7904/1728/3072/8704/19968/69632	4.0×10^5

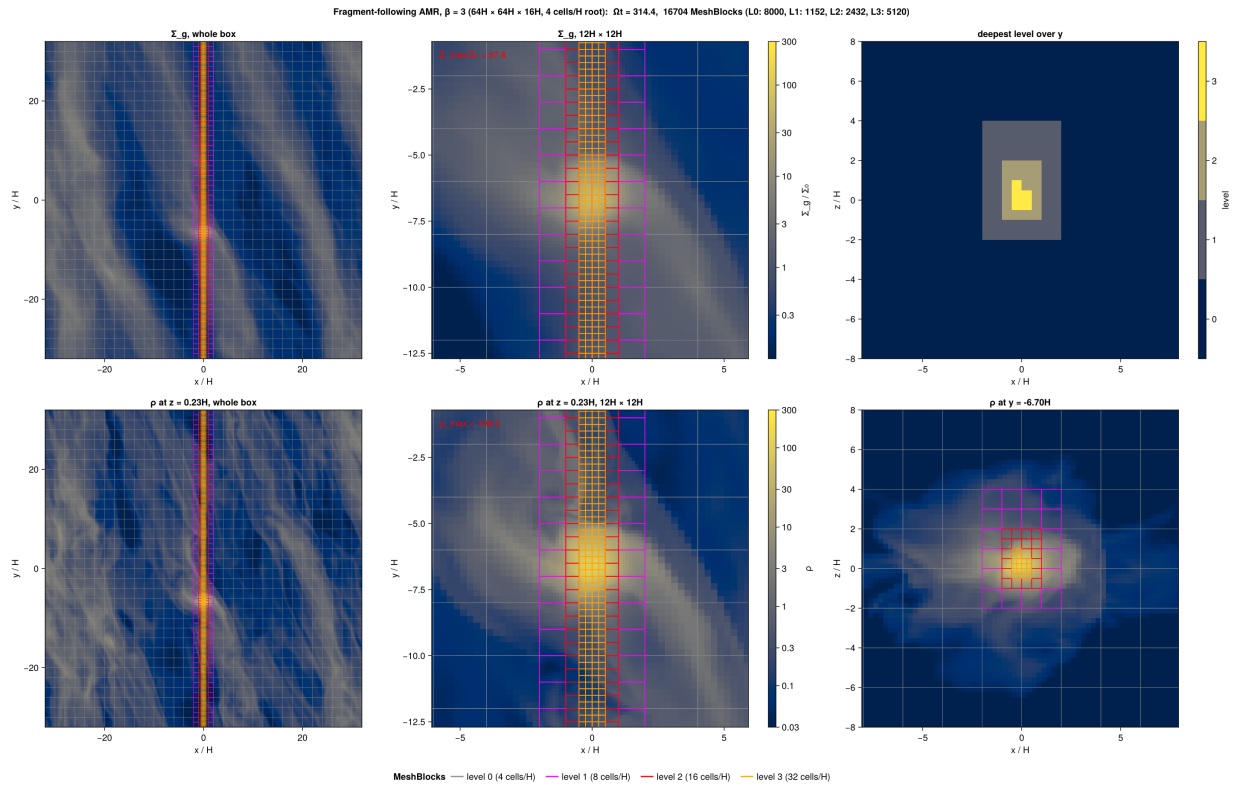


Figure 10: $t = 314.4$, just after the hold: Σ_g with the x–y footprints of all MeshBlocks, and density slices through the maximum with the blocks they cut (gray: root; magenta: 1; red: 2; orange: 3).

- The streams refine to level 1 at most and derefine again; the cascade of the first attempt does not happen.
- The fragment forms at $t \approx 314$, as in the uniform run (≈ 313); the hold takes it to level 3 within one dump ($0.1/\Omega$) of $\rho_{\max}$ passing 100.
- **Runaway contraction.** From $t \approx 316$, $\rho_{\max}$ doubles about every Ω^{-1}. That is far slower than free fall ($\sim 0.01/\Omega$ at $\rho = 10^4$): the fragment is held up by pressure and contracts as it cools, on the $\beta = 3$ cooling time. β cooling has no opacity limit, so the contraction does not stop, and the Jeans criterion adds level 4 at $t \approx 321$ and level 5 at $t \approx 326$.
- **Cost.** About $1/\Omega$ per hour at level 3, $0.19/\Omega$ at level 4 (time step 1.5×10^{-4}, 2.8 s per cycle), $0.08/\Omega$ at level 5 (time step 6×10^{-5}, 1.1×10^5 blocks). The second 24-hour job advanced from $t = 324.9$ to 326.85.
- **Migration.** The fragment moves outward in x: 0 ($t = 314.4$), 3.6 (316.8), 4.8 (319.9), 11.5 (324.9), 10.8 (325.6), and drifts fast in $-y$ with the shear, crossing the y boundary.

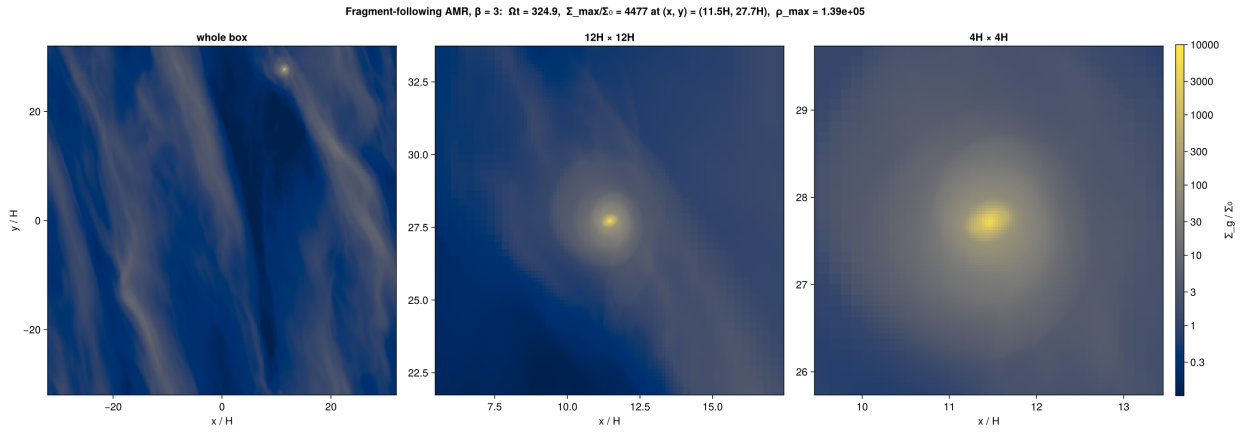


Figure 11: $t = 324.9$: Σ_g/Σ_0 over the whole box and $12H$ and $4H$ around the fragment (map at 32 cells/H, level-4 blocks averaged).

9 The fragment's structure: uniform grid against AMR

Both runs restart from the same $\beta = 10$ state at $t = 300$ but have diverged by $t = 325$, so their fragments are at different places; their structure is compared about each density maximum, in cylindrical R about the fragment's spin axis (the angular momentum of the gas within $1H$), with velocities relative to the fragment and the background shear restored.

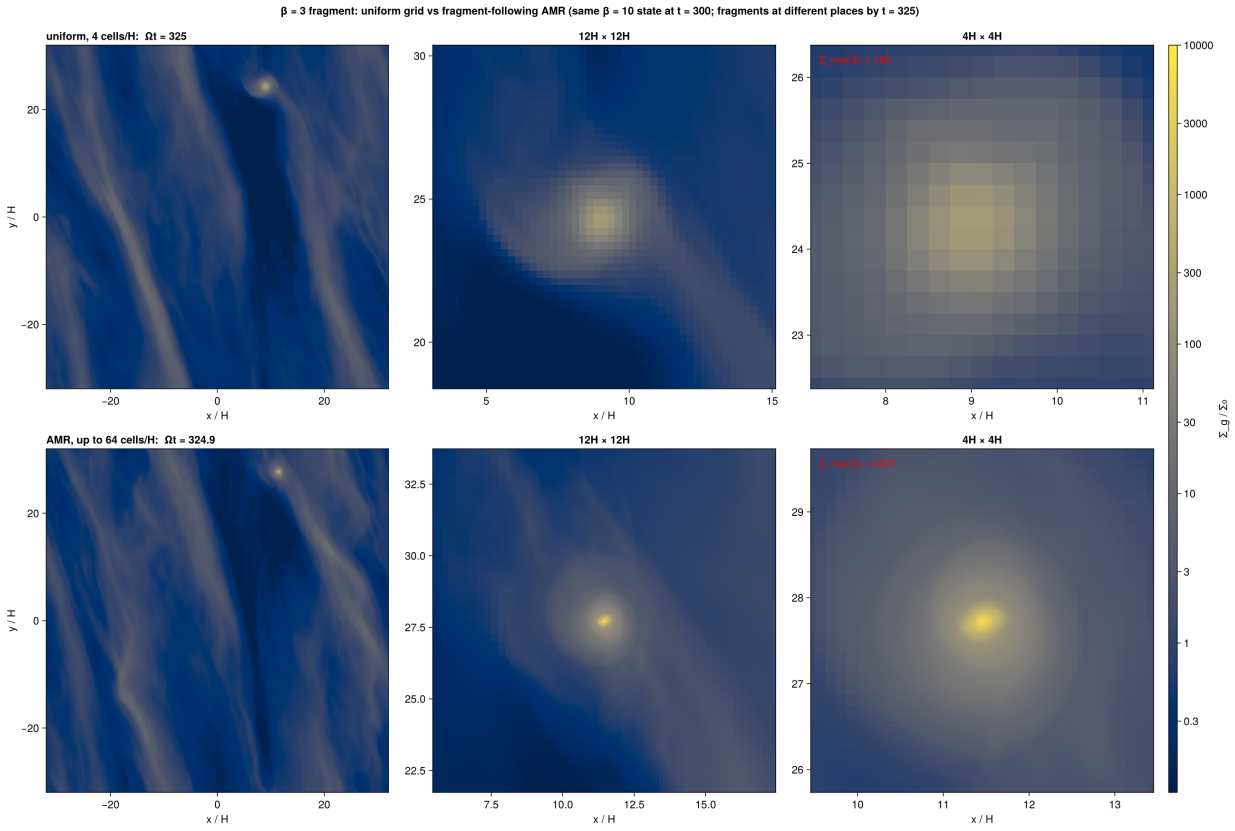


Figure 12: Σ_g/Σ_0 at $t \approx 325$: uniform 4 cells/H (top), AMR (bottom); whole box, $12H$ and $4H$.

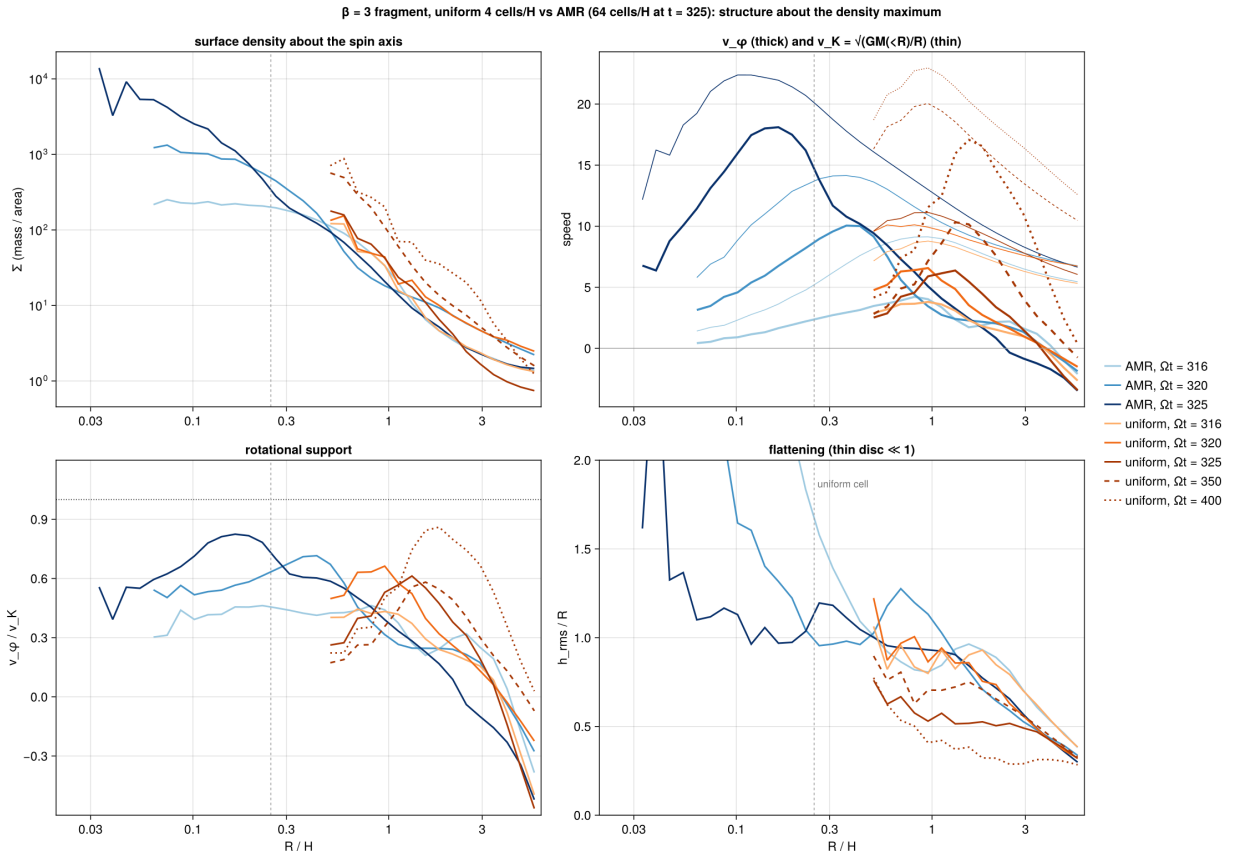


Figure 13: Radial structure: $\Sigma(R)$; v_ϕ (thick) and $v_K = \sqrt{GM(<R)/R}$ (thin); v_ϕ/v_K; $h_{\rm rms}/R$. AMR at $t = 316$, 320, 325 (blues); uniform at 316, 320, 325 (oranges), 350 (dashed) and 400 (dotted). Points within two cells of the finest level are dropped.

- **The outside agrees, the inside does not.** The fragment holds ~ 500 mass units within $2H$ in both runs (6% of the box; ≈ 490 uniform, 555 AMR at $t \approx 325$), and $\Sigma(R)$ nearly coincides beyond $R \approx 0.5H$. Within $0.25H$: 41 (uniform) against 299 (AMR). $\Sigma_{\rm max}/\Sigma_0$: 188 against 4477.

- **Spin-up.** The AMR fragment's peak v_ϕ/v_K rises 0.45 ($t = 316$, at $R \approx 0.6H$) $\rightarrow$ 0.72 (320, $0.35H$) $\rightarrow$ 0.82 (325, $0.15H$): it moves inward and up as the fragment contracts, as angular-momentum conservation demands.

- **Not yet a disc.** $c_s \approx 10$ against $v_\phi \approx 17$, so pressure supplies about a third of the support ($(v_\phi/v_K)^2 \approx 0.67$); $v_R \approx 0$ inside $0.5H$ (quasi-static), infall at 1–4 outside $1H$ (accretion).

- **The uniform grid at $t = 400$:** a pressure-supported core inside $R \approx 1H$ ($c_s = 10$–$15 > v_\phi$; $v_\phi/v_K = 0.2$–0.5) inside a thick, rotating, flattened structure at $R \approx 1.3$–$3H$ ($v_\phi/v_K = 0.75$–0.86, $h_{\rm rms}/R \approx 0.3$–$0.4$); $M(<1H) = 1485$, $M(<2H) = 2093$. It is only 5–12 cells across; its core cannot shrink below a few cells.

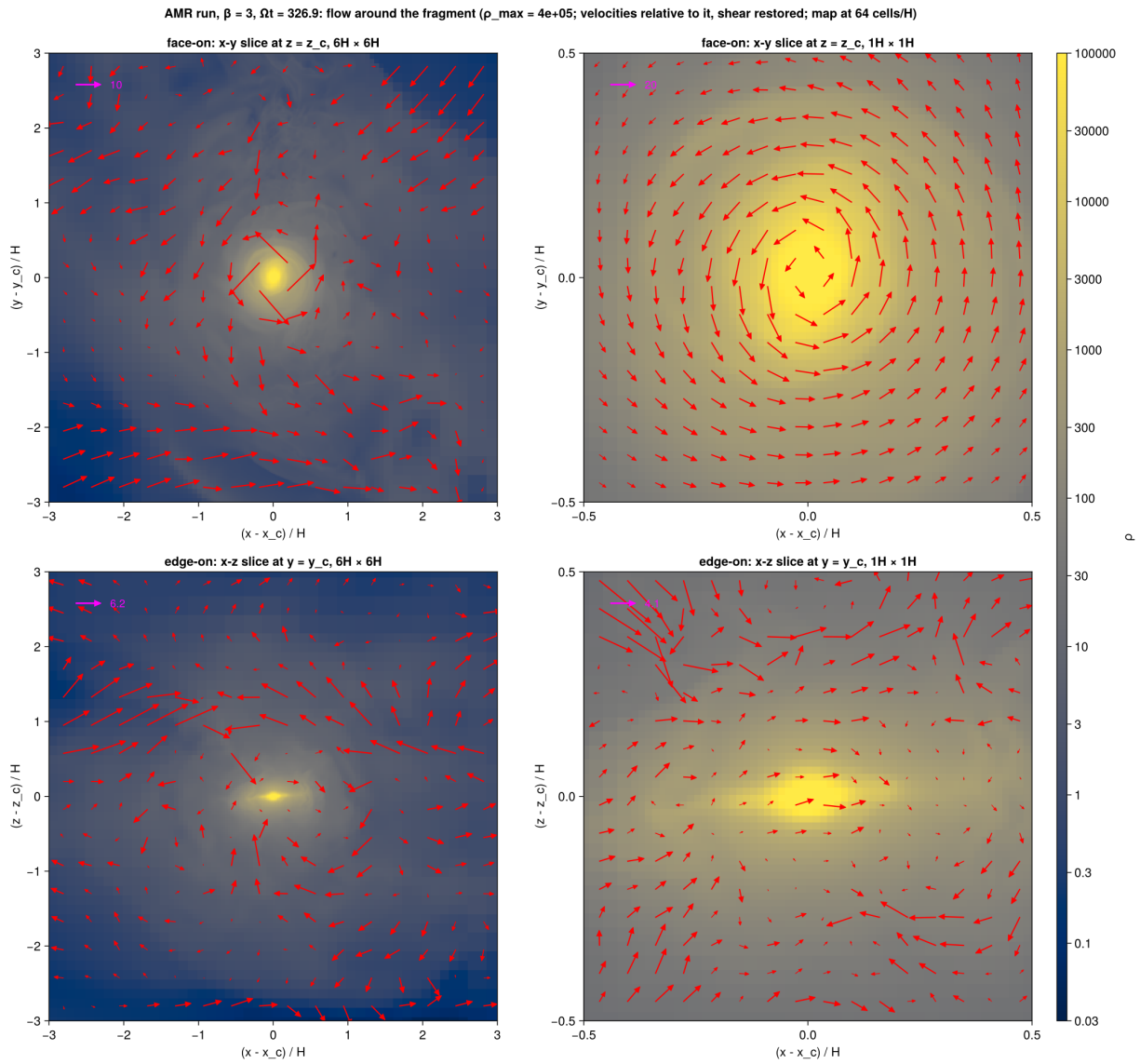


Figure 14: AMR at $t = 326.85$ (the last dump before the first-core restart): density slices through the maximum with velocity vectors relative to the fragment (shear restored), face-on (x–y) and edge-on (x–z), $6H$ and $1H$ windows; the key gives each panel's 90th-percentile speed.

In the $1H$ face-on window the gas rotates prograde at ~ 20 (the panel's 90th-percentile speed) about a slightly elongated core. Edge-on, the densest gas is a layer about $0.4H \times 0.1H$ ($\sim$ 4:1) inside a rounder envelope, and the motions in the x–z plane are slow (~ 4, mostly infall from above) against the rotation: the core spins, it does not fall in. (At $t = 324.9$ the same slices gave rotation ~ 18, poloidal speeds ~ 2 and a $0.5H \times 0.1H$ layer.) In the $6H$ windows rotation dominates within $\sim 1H$; outside, the shear takes over, and gas falls onto the fragment at ~ 6 from the streams above and below the midplane. ($h_{\rm rms}/R$ in the profile figure counts the envelope above and below the core and reads ≈ 1; the slice shows the flattening the profile hides.)

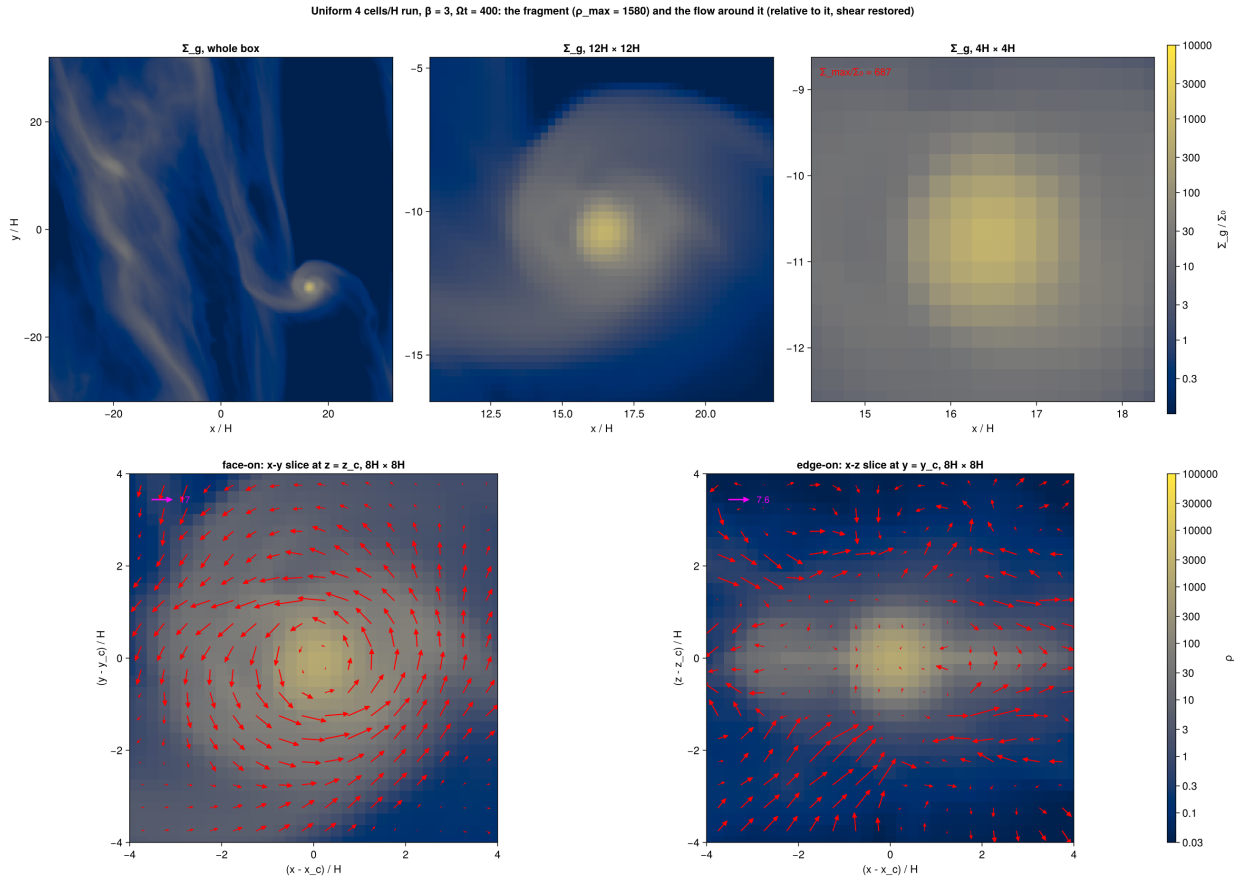


Figure 15: Uniform run at $t = 400$: Σ_g (whole box, $12H$, $4H$) and density slices with velocity vectors, $8H$ windows.

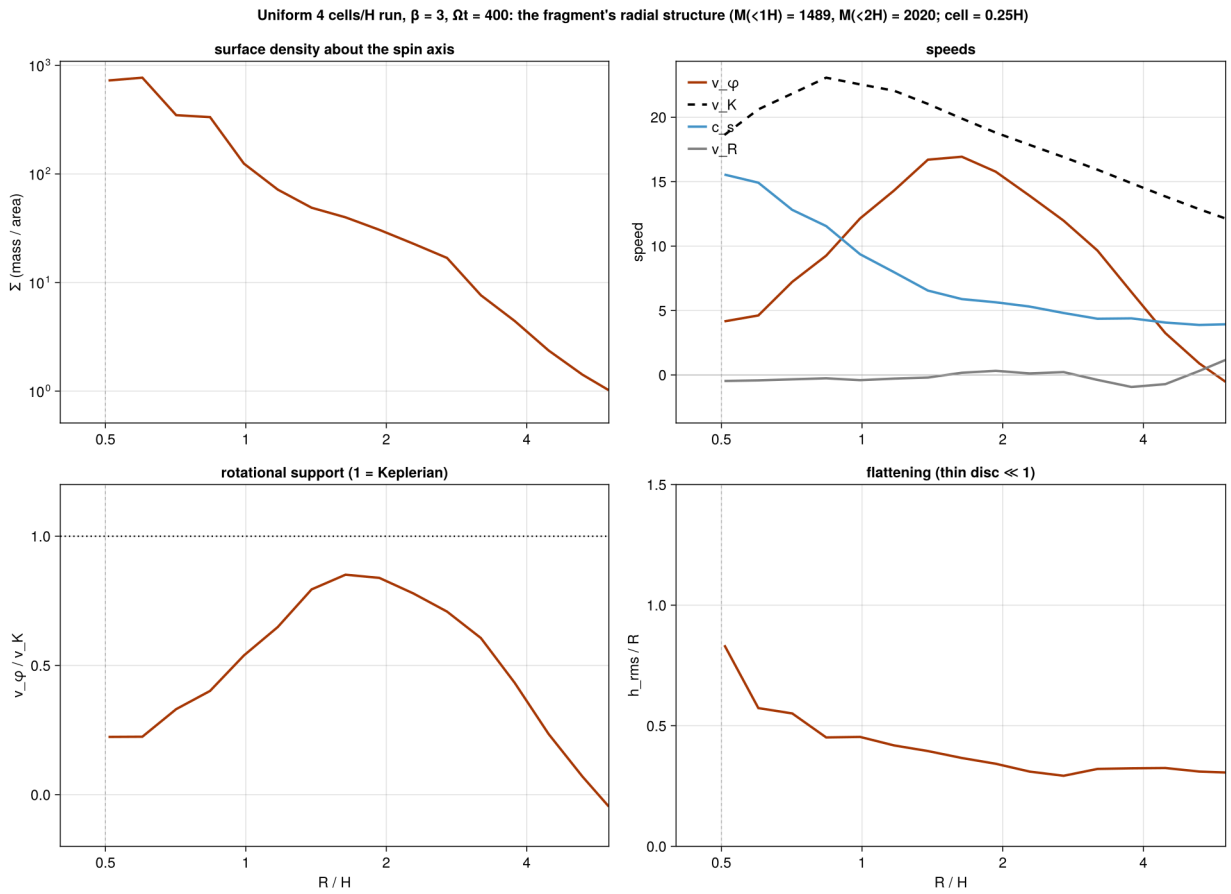


Figure 16: Uniform run at $t = 400$: radial profiles of Σ; v_ϕ, v_K, c_s, v_R; v_ϕ/v_K; $h_{\rm rms}/R$.

10 The circumplanetary disc

Following Szulágyi, Mayer & Quinn (2017), the disc is the gas rotating at $\geq 45\%$ of Keplerian. For each dump (AMR every $0.5/\Omega$, uniform every $1/\Omega$ from $t = 313$): the contiguous band $[R_{\rm in}, R_{\rm out}]$ around the peak of v_ϕ/v_K where it is ≥ 0.45; $M_{\rm CPD}$, the mass of the cells in that band whose own $v_\phi \geq 0.45\, v_K$; $M_{\rm core} = M(< R_{\rm in})$; the Hill radius $r_H = [GM(< r_H)/3\Omega^2]^{1/3}$.

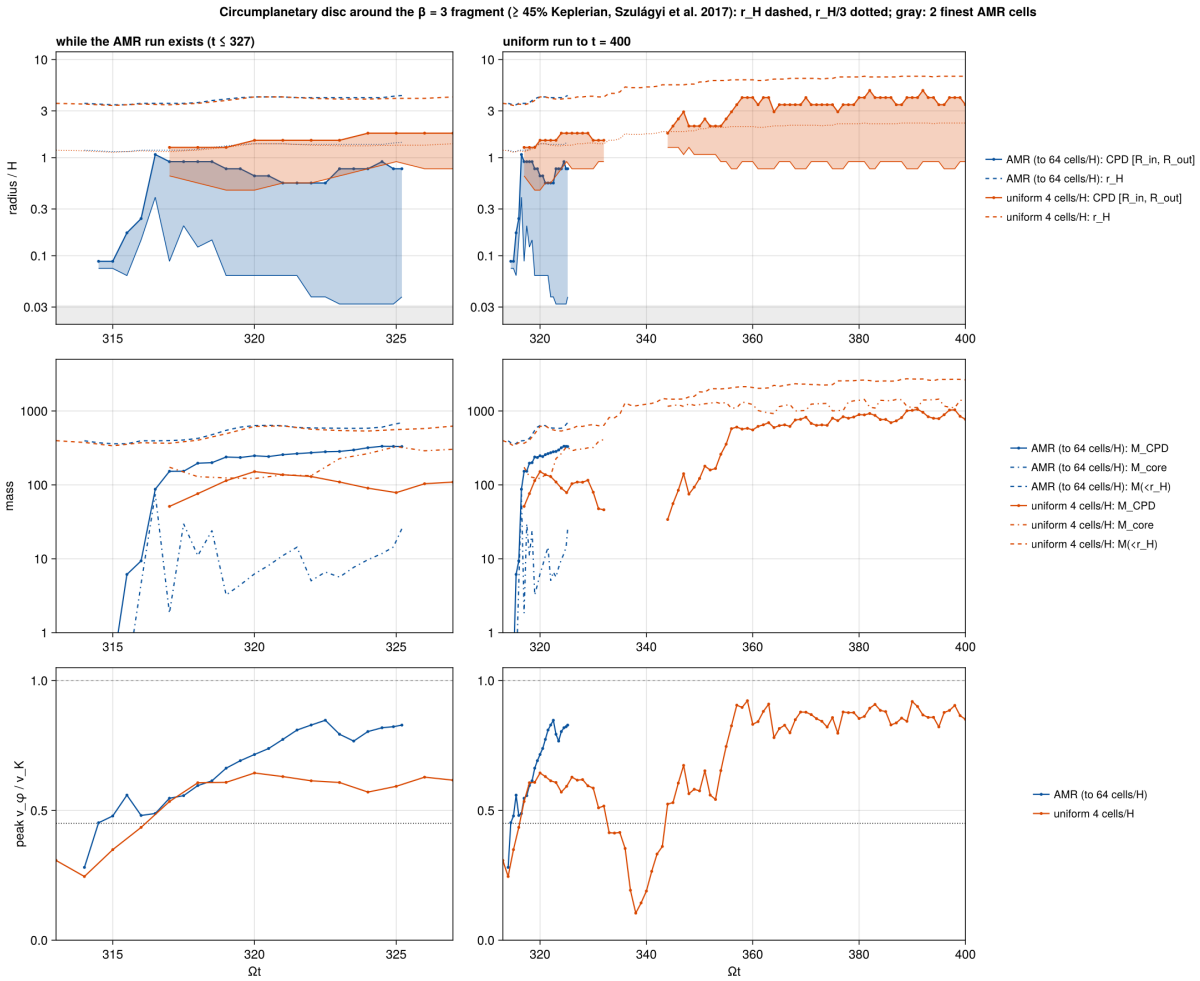


Figure 17: Disc radius band and Hill radius (top), disc, core and Hill masses (middle), peak v_ϕ/v_K (bottom); AMR blue, uniform orange. Left: while the AMR run exists; right: the uniform run to $t = 400$. Gray band: two cells of the finest AMR level.

- **AMR** ($t = 316.5$–325.2): the disc appears $\sim 2/\Omega$ after the fragment. $R_{\rm out}$ settles at 0.55–1H ($\approx 0.8H$, $0.2\,r_H$, at $t = 325$); $R_{\rm in}$ sits at two finest cells ($\approx 0.03H$) — the gas rotates above 45% Keplerian down to the resolution limit, so no slowly rotating core is resolved ($M_{\rm core} < 26$). $M_{\rm CPD}$ grows $150 \to 330$, half of $M(< r_H) \approx 660$.
- **Uniform**: $t = 317$–332, disc at 0.5–1.8H with 50–150 against a core of 120–330 (core-dominated); destroyed at $t = 333$–343 (the outward jump of sec. 6); re-formed from $t = 344$, and at $t = 360$–400 it spans 0.8–4.1H ($R_{\rm out} \approx 0.6\,r_H$) with $M_{\rm CPD} = 600$–1050 against a core of 1000–1400 and $M(< r_H) = 2000$–2700 (30–40%); peak $v_\phi/v_K = 0.85$–0.92.

Against the literature. Disc-to-planet mass ratios for fragments of gravitational instability are $\sim 25\%$ (Shabram & Boley 2013) to $\sim 50\%$ (Galvagni et al. 2012), as summarized by Szulágyi et al. (2017), whose own $10M_J$ clump peaks at $v_\phi/v_K \approx 0.7$ with a disc at 2–6 AU. Our ratios (AMR $\approx 50\%$, late uniform 30–40%) are in that range, and our rotation profiles have their shape (slow core, sub-Keplerian peak, fall-off into the shear). Refinement changes how the mass divides: at $t \approx 325$ the uniform fragment is core-dominated (80 in the disc, 330 in the core), the AMR fragment rotation-dominated. The 45% criterion does not require the gas to be thin; in the AMR run it counts the thick rotating interior as disc.

11 The first-core run (in progress)

`b3_renorm_clumpamr_fc` restarts the fragment-following run from its $t = 324.94$ restart file (levels ≤ 4, $\rho_{\max} \approx 1.4 \times 10^5$) with `bcool_rho_cut` $= 10^5$ (cooling $\times\, 1/[1 + (\rho/10^5)^4]$) and the deepest level capped at 4 (64 cells/H), to $t = 335$. The restart read the overlay (5 levels, 54 784 blocks, mass target 8202.52) and its first steps reproduce the old run's to the digits the cutoff does not touch (2.8 s per cycle, time step 1.4–1.5×10^{-4}). The test is whether $\rho_{\max}$ levels off at a few $\times 10^5$ and the time step stops shrinking; then whether v_ϕ/v_K reaches 1 in the inner 0.1–0.2H and that gas flattens. Results will be added here.

12 Summary and open issues

1. The multigrid force error at a level jump drifts a clump toward the coarse side at $\sim 2 \times 10^{-3}$ of its surface gravity; 8 fine cells of buffer remove it.

2. In the $64H \times 64H \times 16H$ box renormalization holds the mass without changing Q; WENO-Z and plm give the same $\beta = 10$ turbulence.

3. The stress balances β cooling (SC14 eq. 21) with WENO-Z+FOFC from $\beta = 4$ to 80 (1.01–1.07), the energy budget closing with the boundary outflow; plm falls to 0.82 at $\beta = 80$ (§5).

4. $\beta = 10$: no fragment (uniform to 300, Jeans AMR to 1000). $\beta = 5$: none in $110/\Omega$ so far. $\beta = 3$: WENO-Z fragments at $t \approx 313$; plm does not — at 4 cells/H the reconstruction decides.

5. A followed fragment contracts on the cooling time without limit under β cooling; the cost of following it is set by the deepest level's time step. The first-core cutoff is the remedy under test.

6. With resolution the fragment becomes compact and rotation-dominated (peak v_ϕ/v_K 0.83, flattened core); its disc by the 45% criterion holds half its Hill mass, consistent with the published 25–50%.

7. Open: the heap corruption at $\sim 10^5$ blocks; the multigrid interface error; fragmentation itself is decided on the 4 cells/H root grid (the boundary converges only above ~ 40 cells/H, Baehr et al. 2017); no magnetic fields (Kubli et al. 2023 find much less angular momentum in magnetized clumps).